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CAPSTONE

ACADEMIC SUCCESS AND DROPOUT EXPLORATION

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Introduction

Essential Question

How can an institution identify students who may need support early enough to improve student outcomes?

Project Objective

Predict college dropout rates given a student's academic and socioeconomic background.

Motivation

Education is crucial to society, which makes predicting dropout rates all the more important. The ability to predict these dropout rates allows us to identify students at risk early, which protects both the student's well-being and the university's own funding and enrollment (Delogu, 2024).

Data Overview

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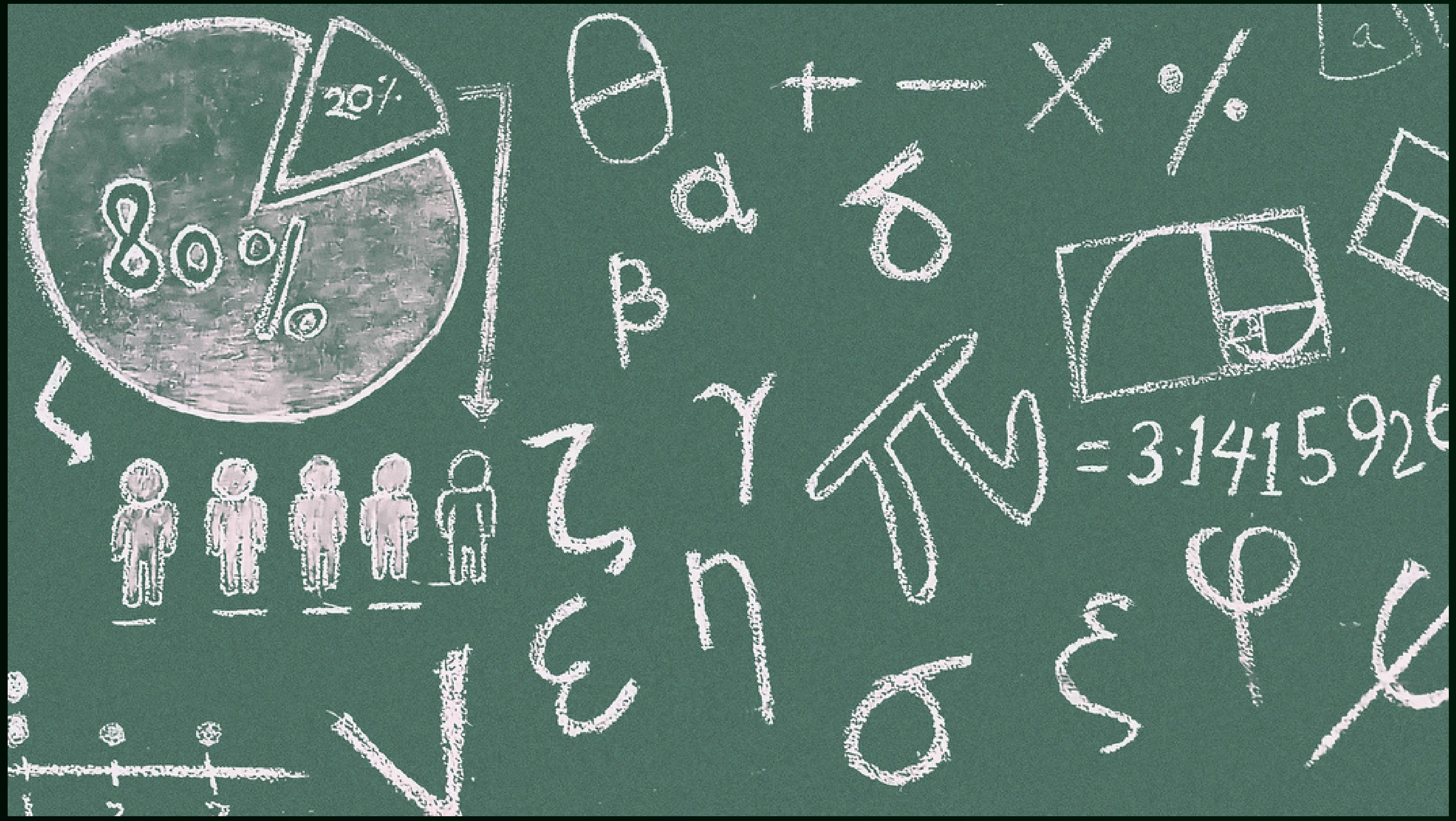
- **36 predictors**
  - **All numerical**
- **4424 rows**
  - **No null or duplicate values**
- **Target variable: Dropout rate**
  - **“Dropout” or “Graduate”**

```
0   Marital Status
1   Application mode
2   Application order
3   Course
4   Daytime/evening attendance
5   Previous qualification
6   Previous qualification (grade)
7   Nacionality
8   Mother's qualification
9   Father's qualification
10  Mother's occupation
11  Father's occupation
12  Admission grade
13  Displaced
14  Educational special needs
15  Debtor
16  Tuition fees up to date
17  Gender
18  Scholarship holder
19  Age at enrollment
20  International
21  Curricular units 1st sem (credited)
22  Curricular units 1st sem (enrolled)
23  Curricular units 1st sem (evaluations)
24  Curricular units 1st sem (approved)
25  Curricular units 1st sem (grade)
26  Curricular units 1st sem (without evaluations)
27  Curricular units 2nd sem (credited)
28  Curricular units 2nd sem (enrolled)
29  Curricular units 2nd sem (evaluations)
30  Curricular units 2nd sem (approved)
31  Curricular units 2nd sem (grade)
32  Curricular units 2nd sem (without evaluations)
33  Unemployment rate
34  Inflation rate
35  GDP
```

Feature Engineering

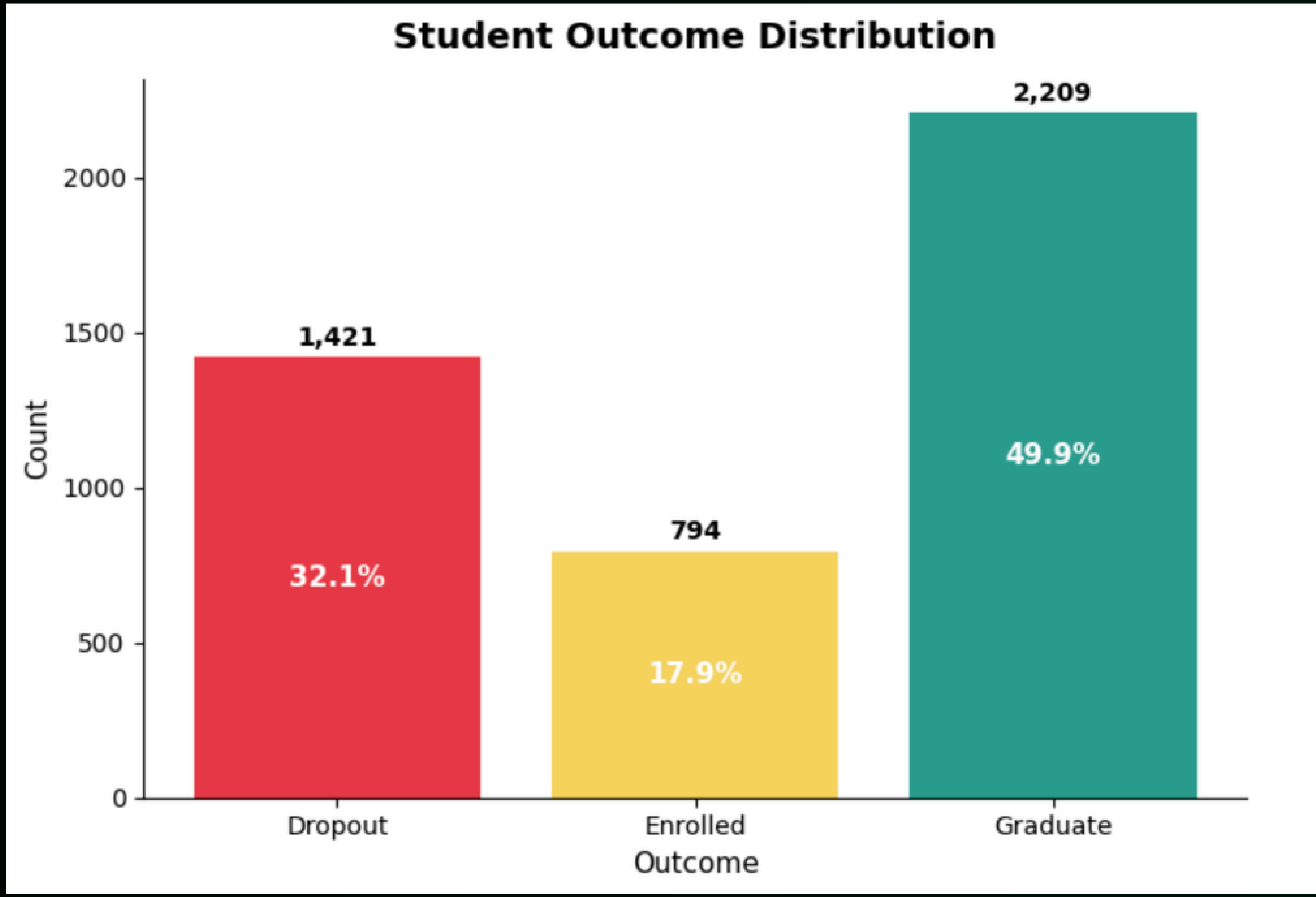
- Approval (passing) rate by semseter
- Overall approval rate
- Grade trend
- Financial risk
- Scholarships
- Scholarship with no debt
- Scholarship with debt
- Age (older students)
- One-hot encoding some of the categorical features expressed as numbers with no ordinal meaning



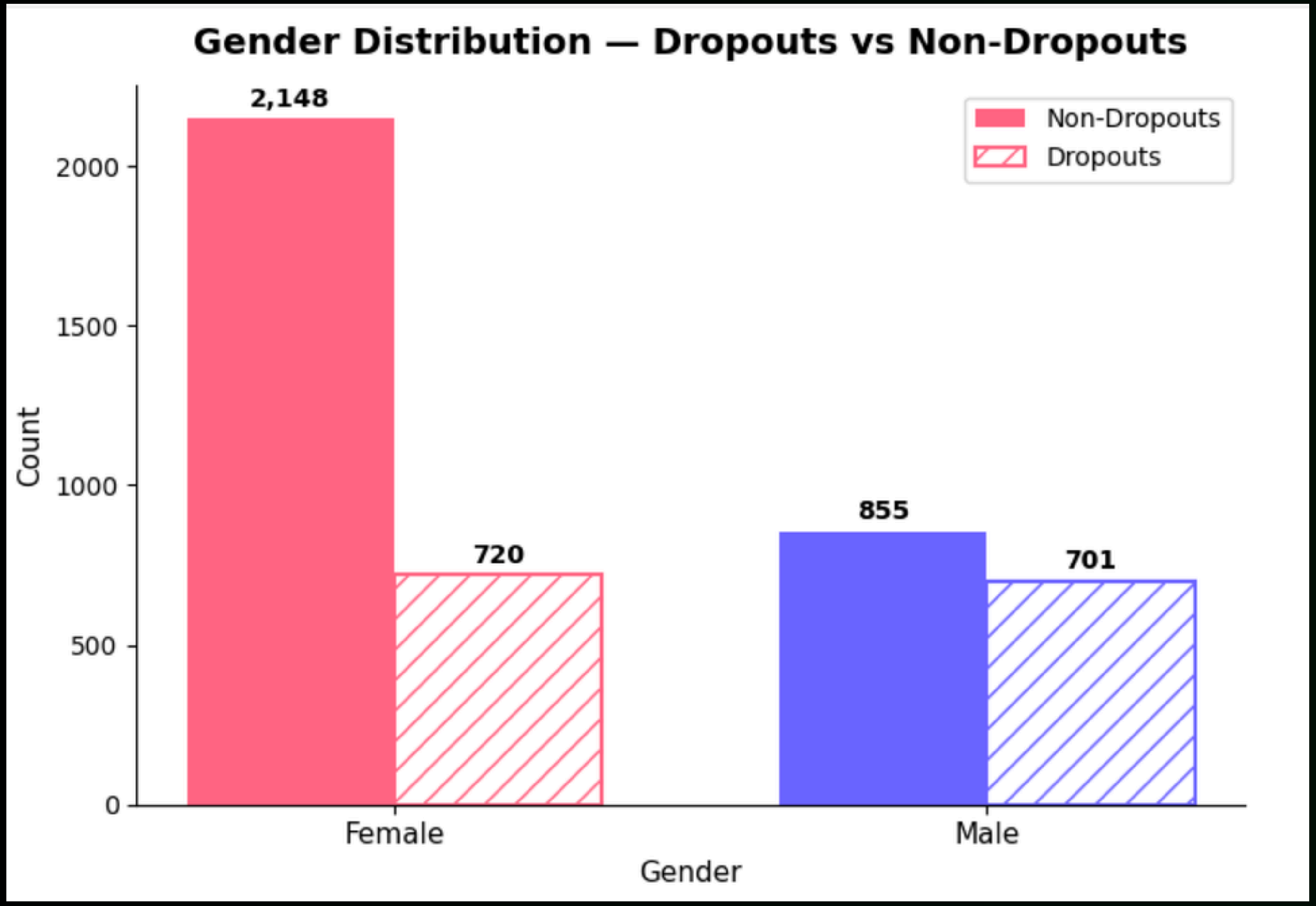
Exploratory Data Analysis

Understanding the Demographics of the Dataset

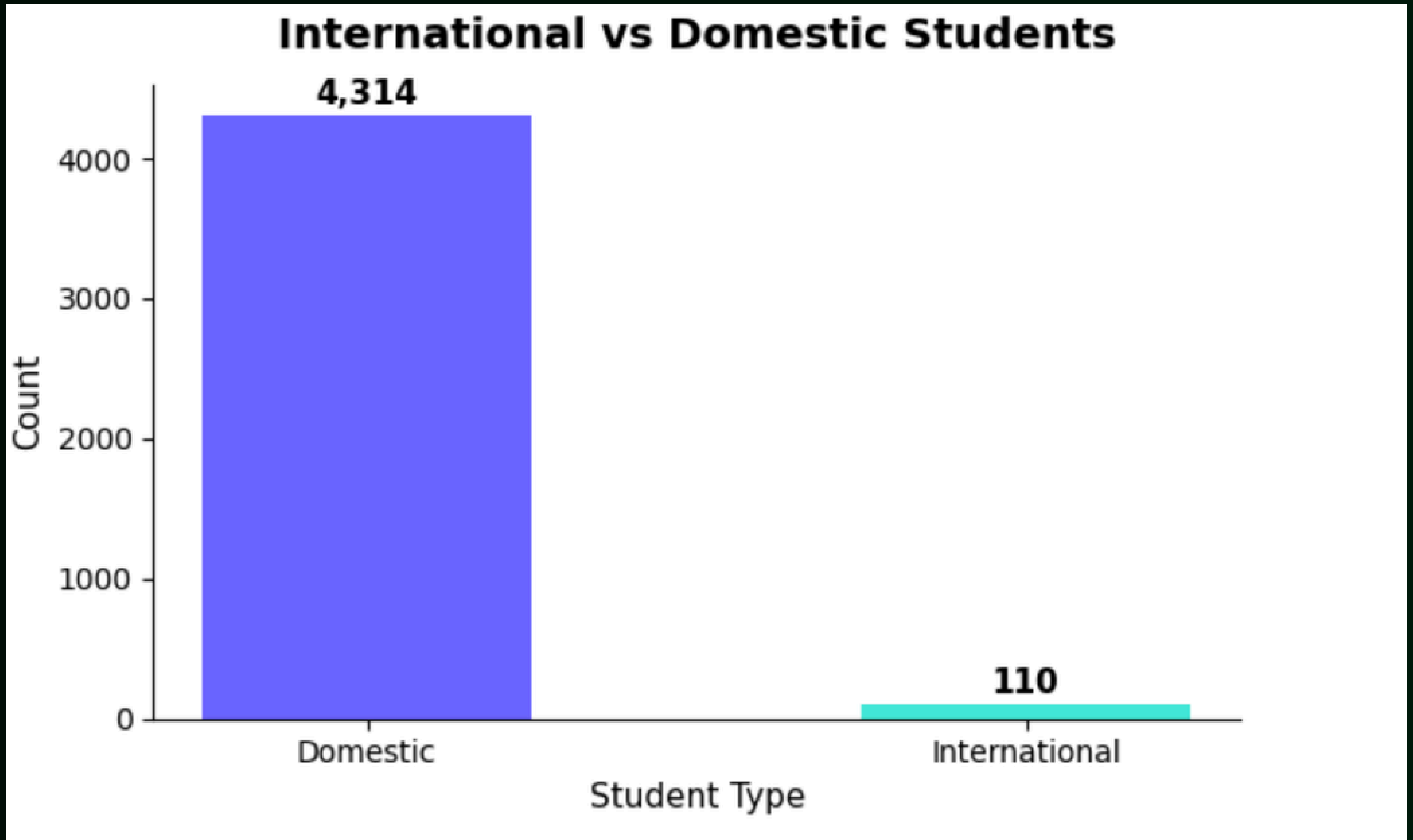
How many students in this dataset dropped out, are currently enrolled, or graduated?



Gender



Origin



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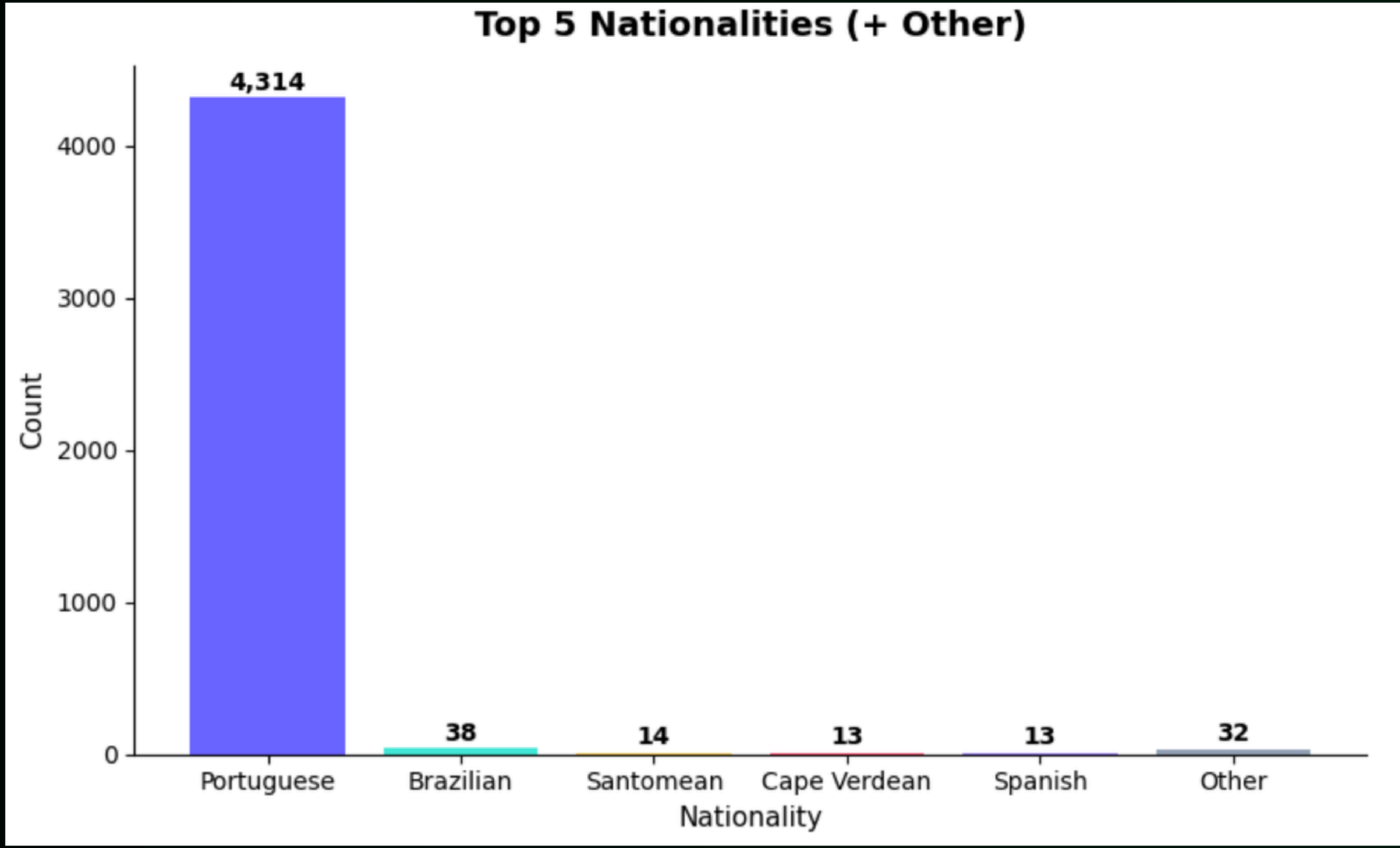
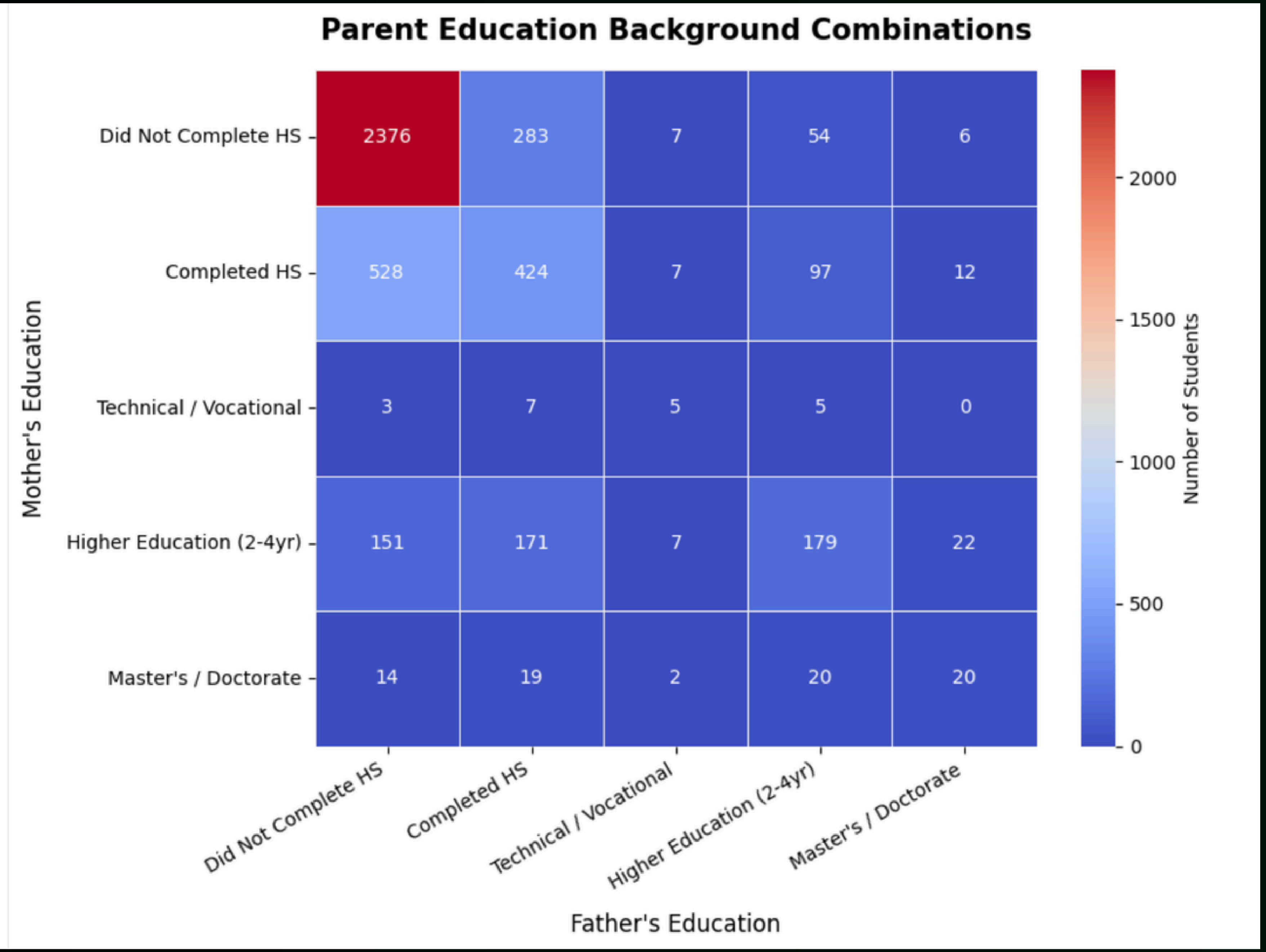
Nationalities of the Students



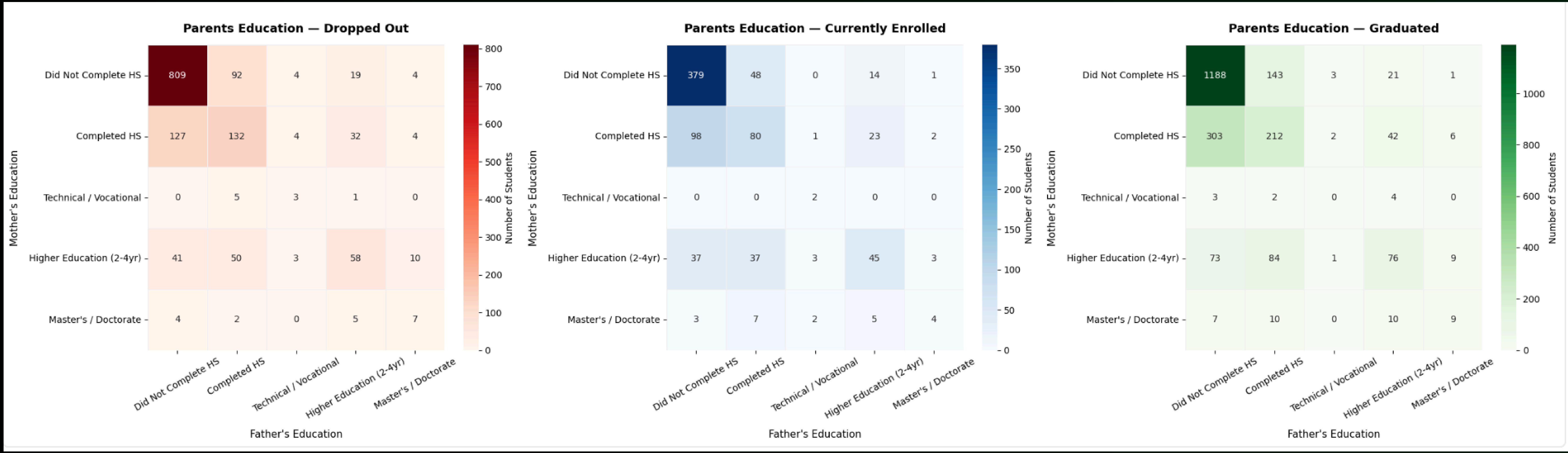
Looking at Parental Backgrounds of Students

Hypothesis: I hypothesized that parental educational attainment might influence a student's college experience and, as a result, could be associated with whether students ultimately graduated, remained enrolled, or dropped out.

Parent Backgrounds



Parent Backgrounds

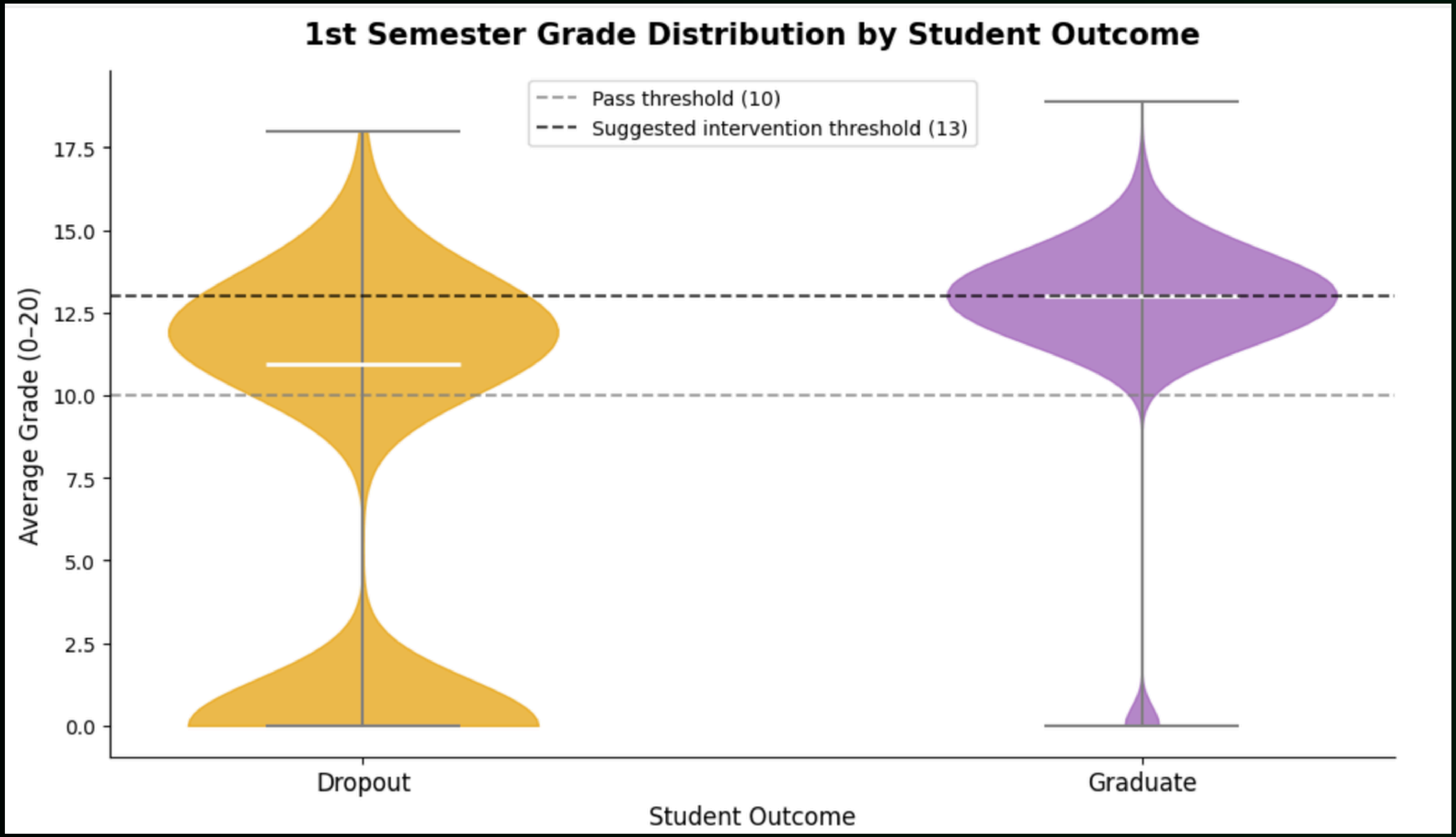


Academic Success and Stuggle

Hypothesis: One of the most intuitive hypotheses was that academic performance would be related to student retention, so I explored whether students who eventually dropped out showed signs of academic struggle during their first semester.

Looking at Academic Struggle

note: the white line indicates the median average grade



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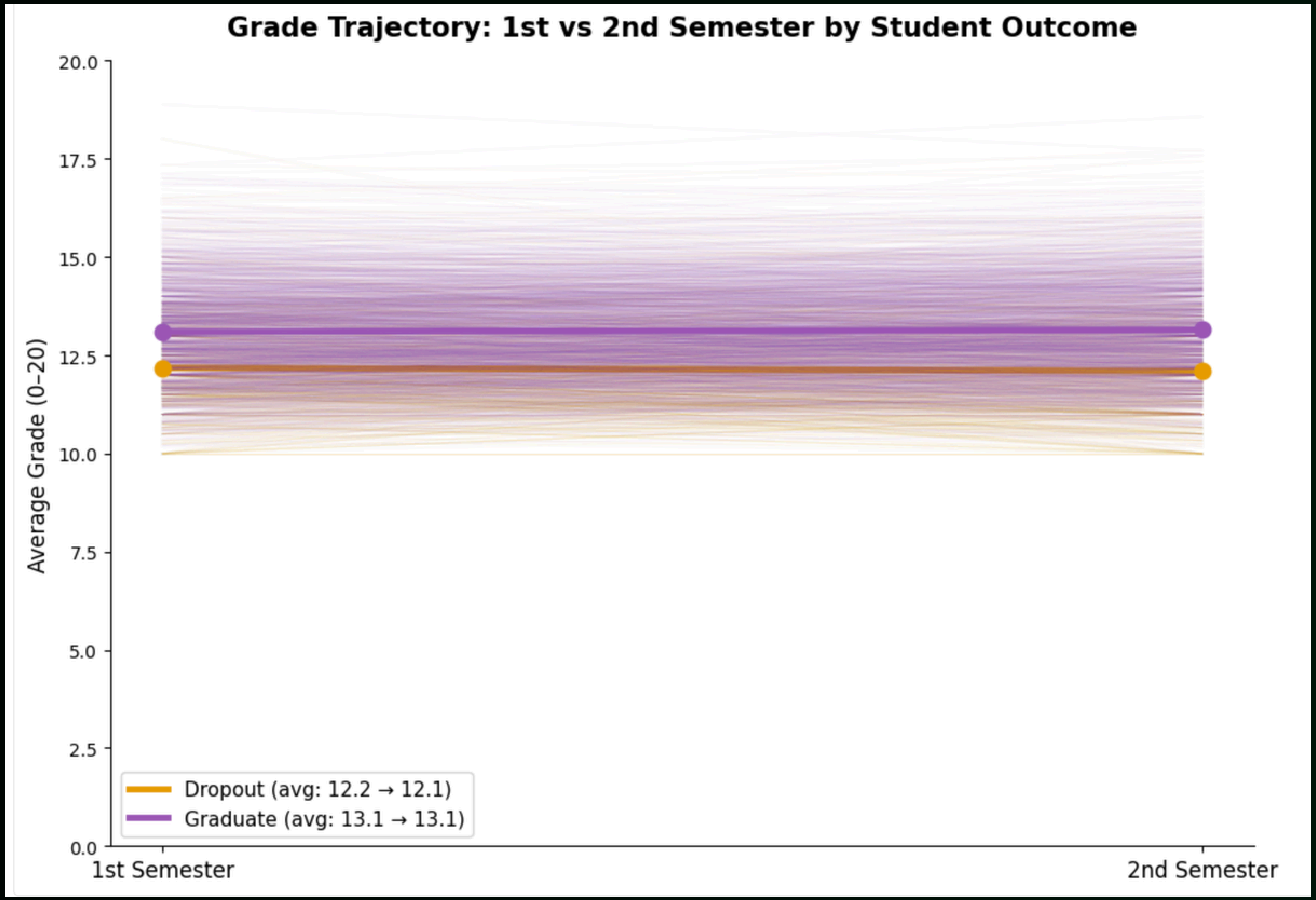
What about 2nd Semester?



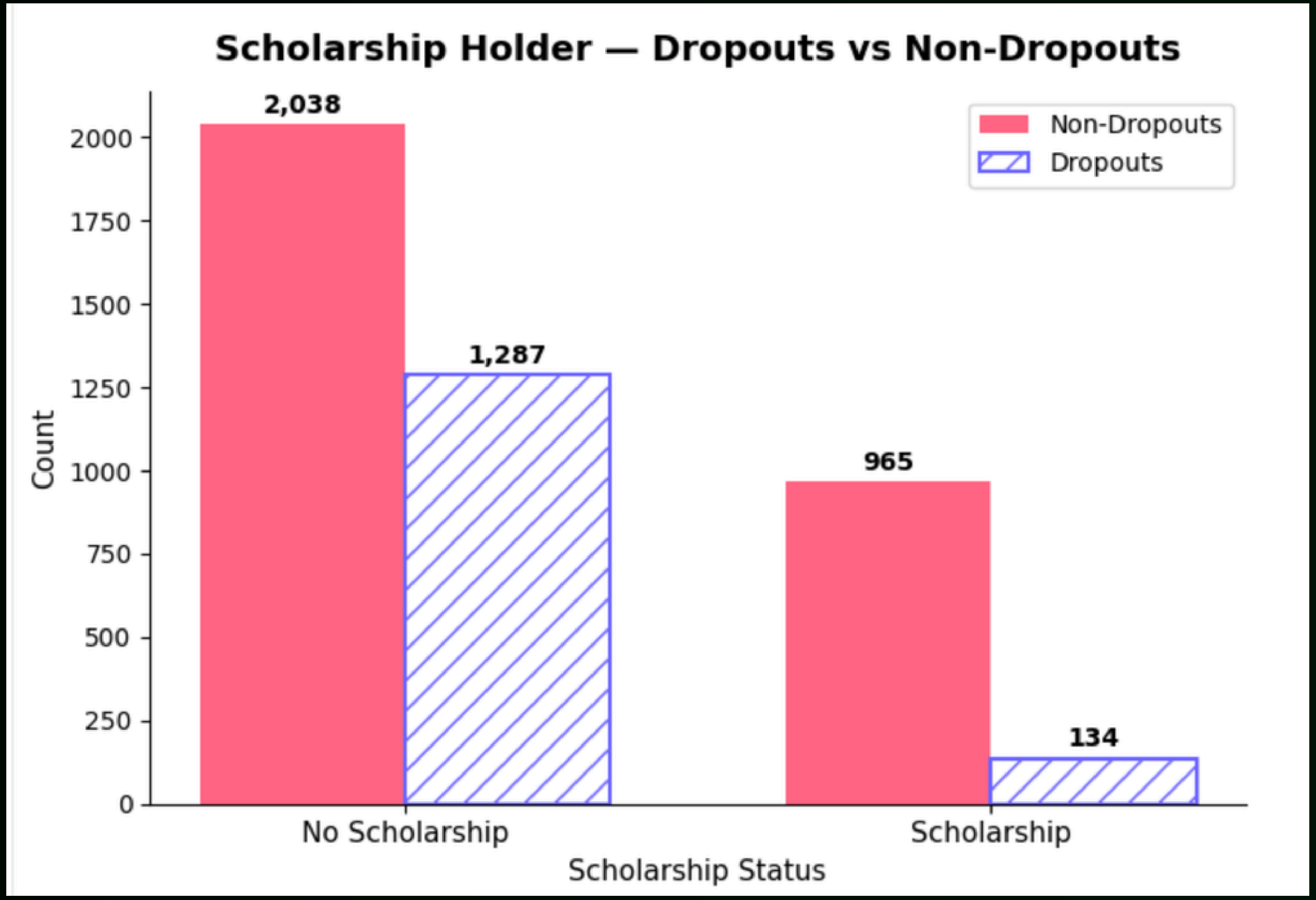
Recommendation #1

Implement an early intervention system for students showing signs of academic struggle.

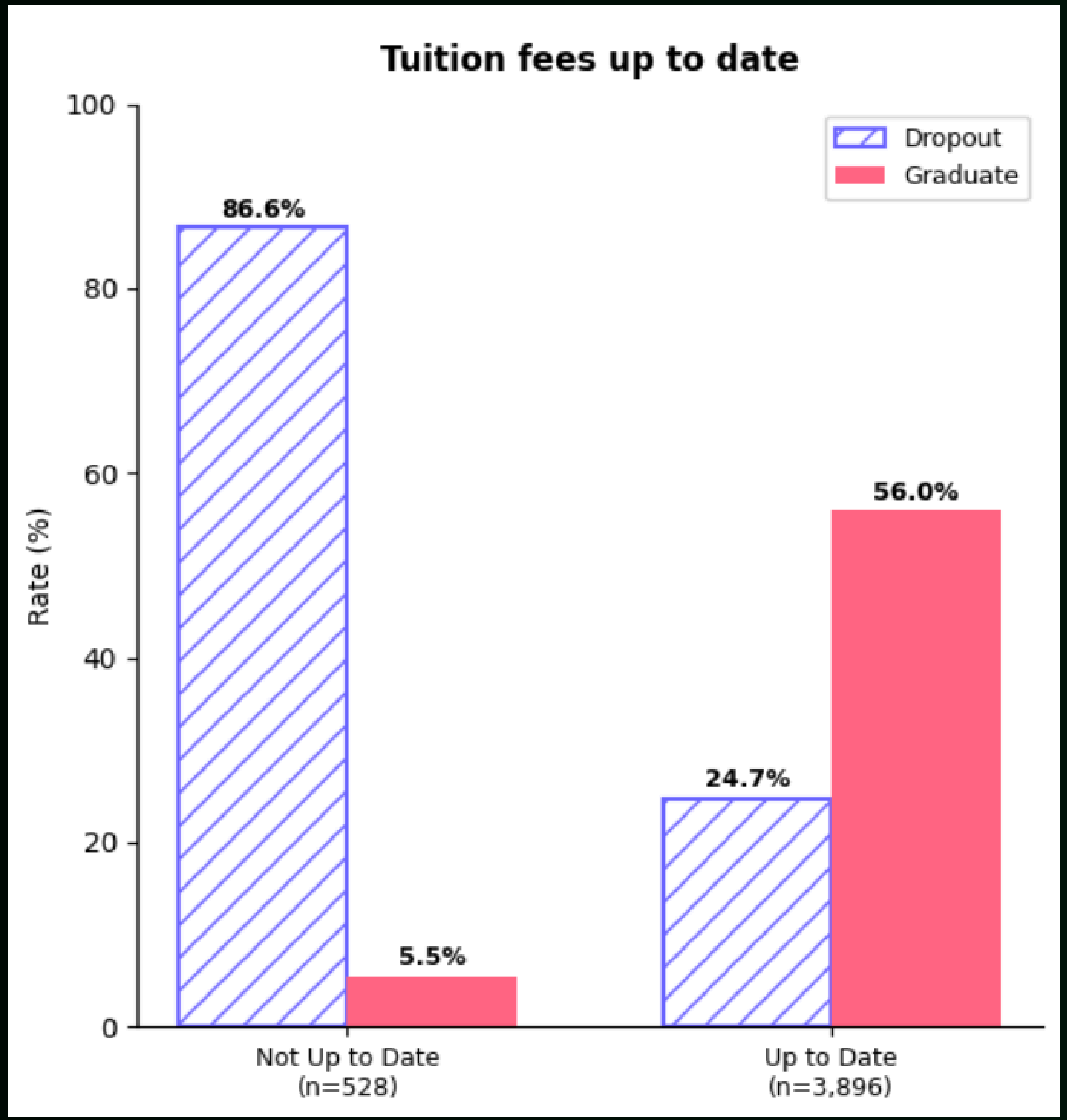
Finances

Hypothesis: Since financial barriers are often cited as a reason students leave higher education, I explored whether students receiving scholarships were more likely to persist and ultimately graduate. Another factor I explored was whether students who were behind on tuition payments were more likely to leave the university than students whose accounts were in good standing.

Scholarship Holders



Finances as an Indicator



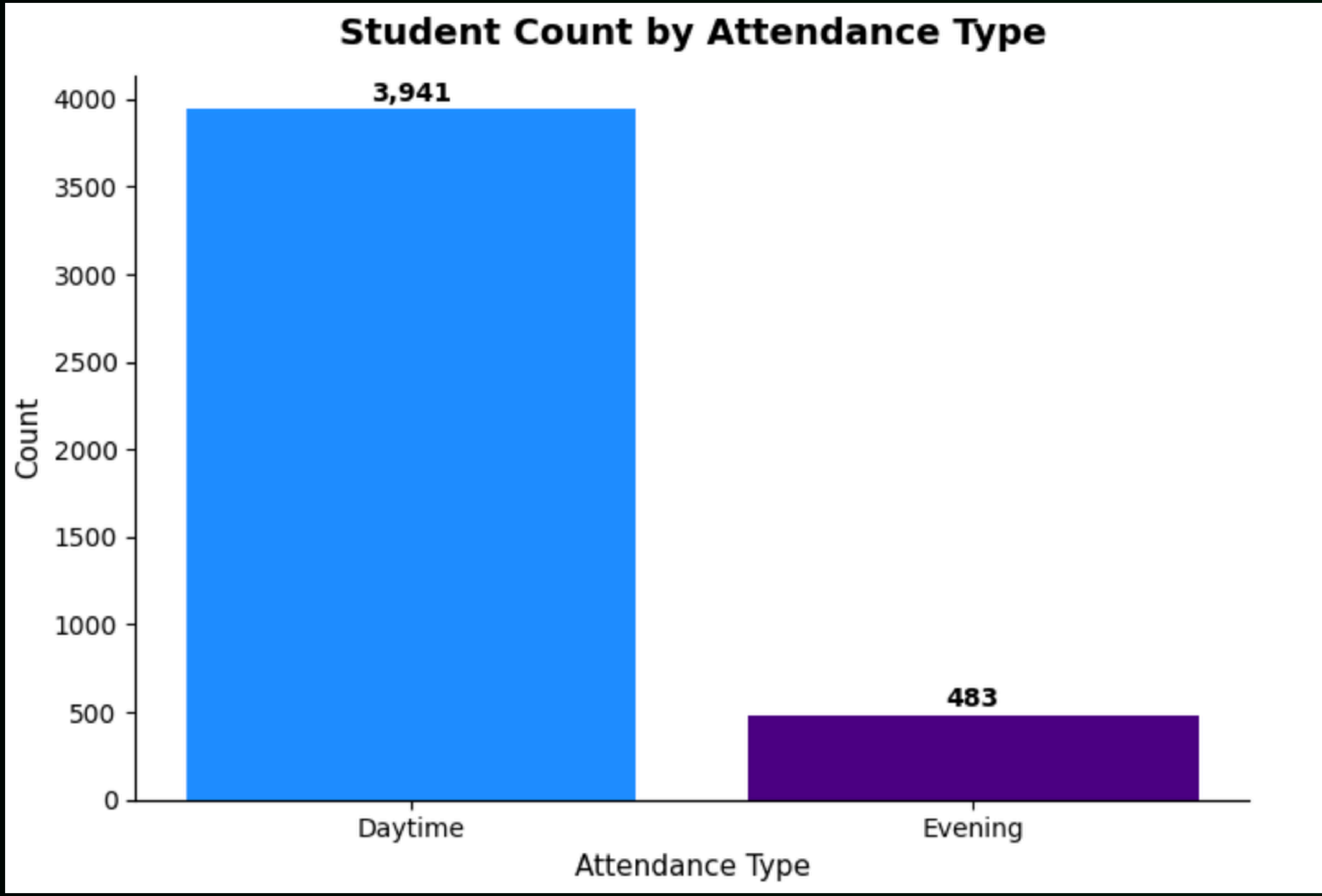
Recommendation #2

Explore expanded financial support opportunities for students at risk of dropping out.

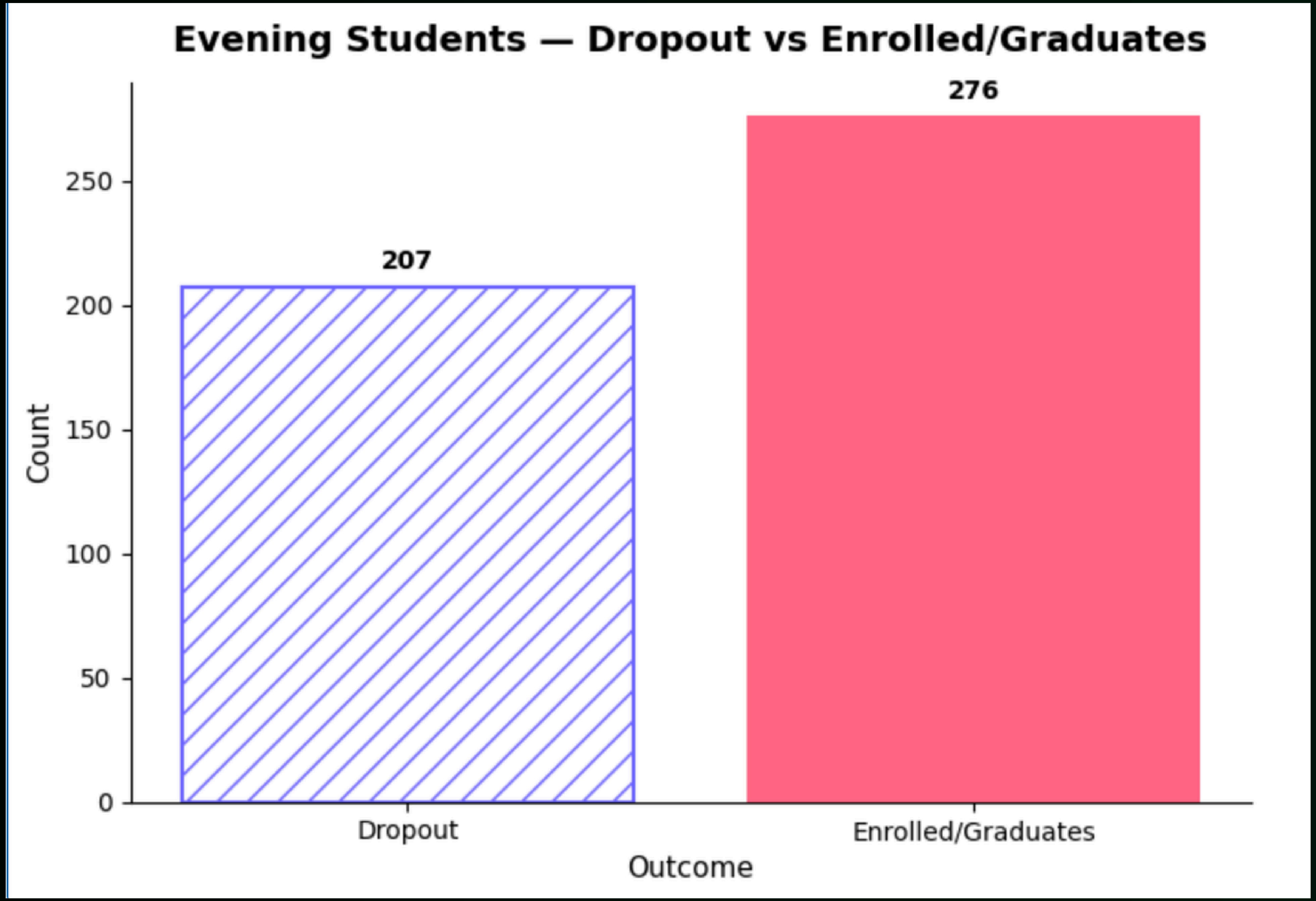
Evening Students

Hypothesis: I hypothesized that enrollment in evening courses might serve as an indicator of different life circumstances and therefore could be associated with student retention outcomes.

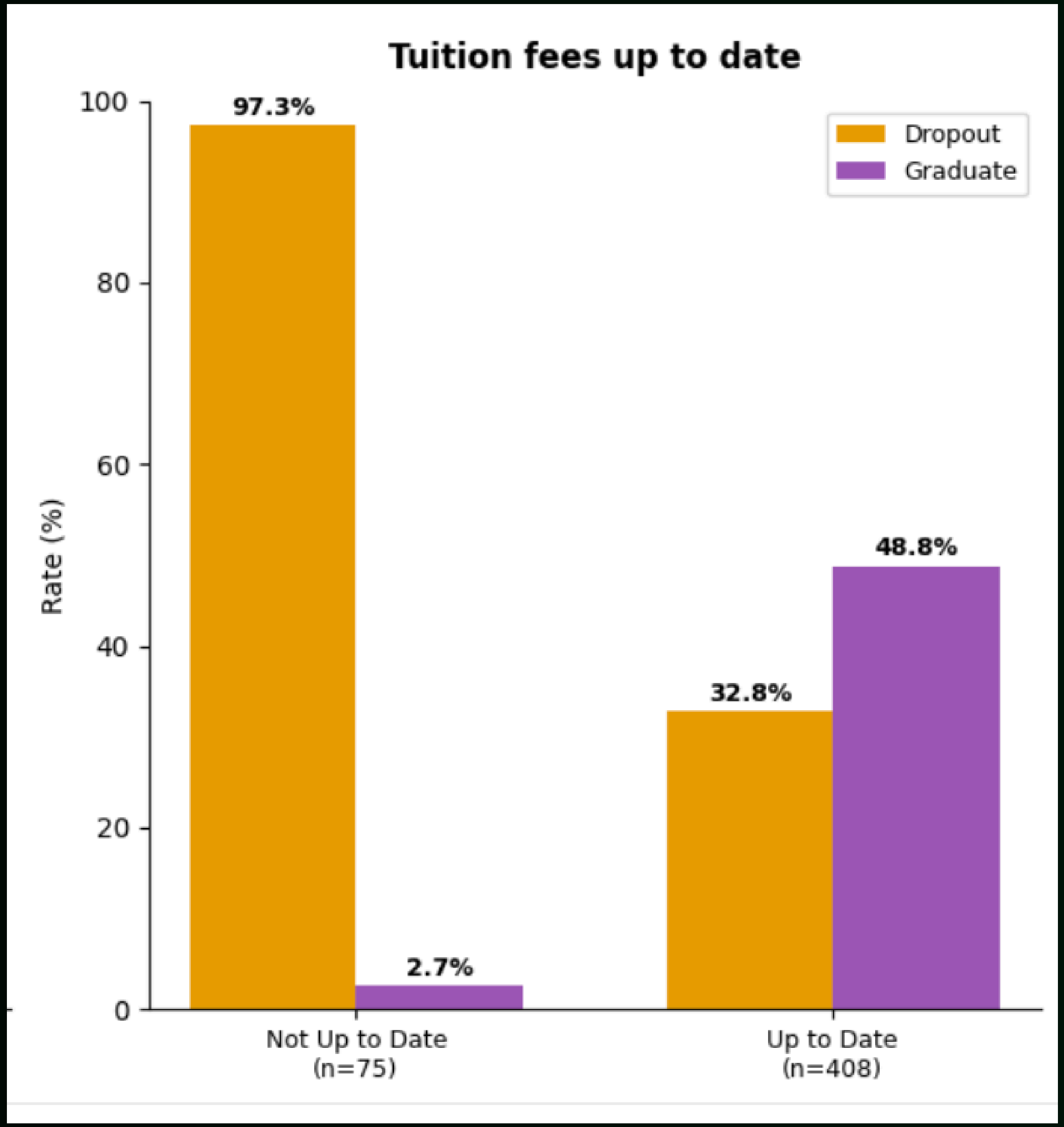
Taking a Closer Look at Evening Students



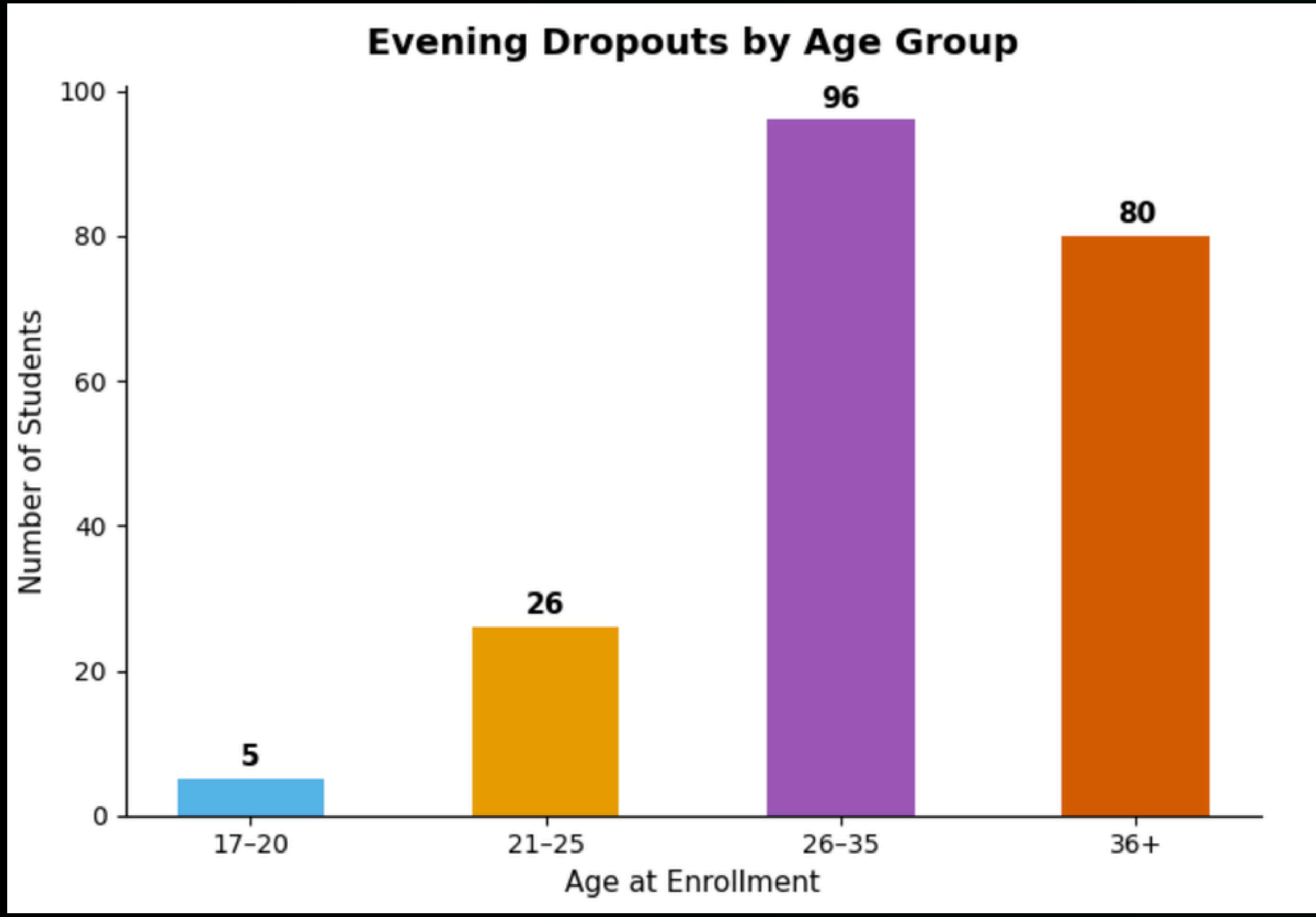
Taking a Closer Look at Evening Students

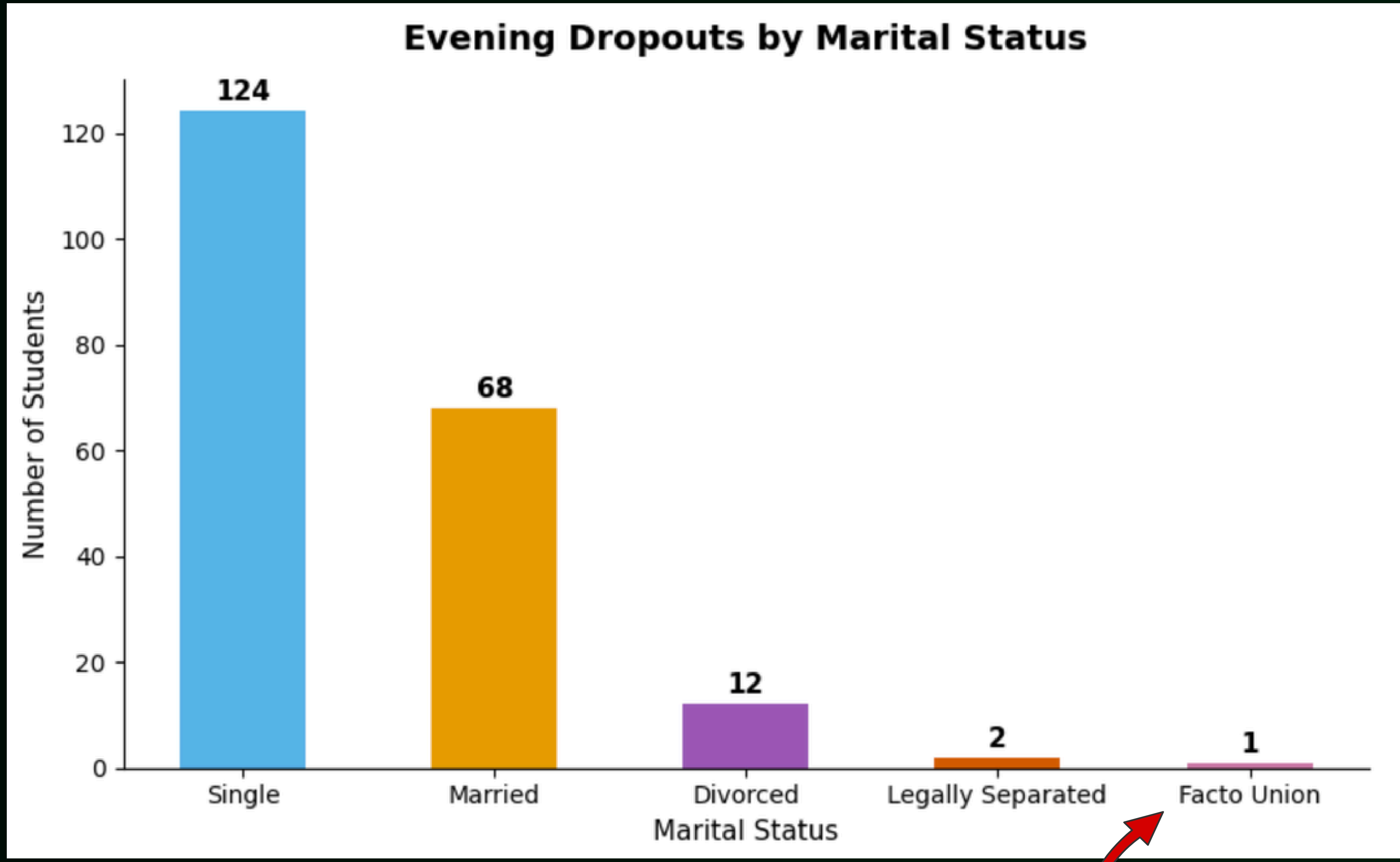


Tuition Fees ~ Evening Students



Taking a Closer Look at Evening Students





facto union is a legal status where two people live together as a couple and share a household but are not legally married

Recommendation #3

Develop targeted support programs for evening and non-traditional students.

Our Predictive Model

Model Performance: Logistic Regression vs. Random Forest

76.9% for Logistic Regression, 77.9% for Random Forest. The real story is in how each model does per outcome, not the overall number. Both models are strong at identifying Graduates (86% and 85% f1-score) and decent at Dropouts (78% and 80% f1-score), but both struggle badly with the "Enrolled" category (only 41% and 46% f1-score). That's worth saying out loud in your presentation: "Enrolled" is the hardest class to predict because it's a moving target — an enrolled student hasn't resolved into success or failure yet, so there's less of a clear signal in the data pointing one way or the other.

=== Logistic Regression ===

Accuracy: 0.769

	precision	recall	f1-score	support
Dropout	0.79	0.77	0.78	284
Enrolled	0.54	0.33	0.41	159
Graduate	0.80	0.93	0.86	442
accuracy			0.77	885
macro avg	0.71	0.68	0.68	885
weighted avg	0.75	0.77	0.75	885

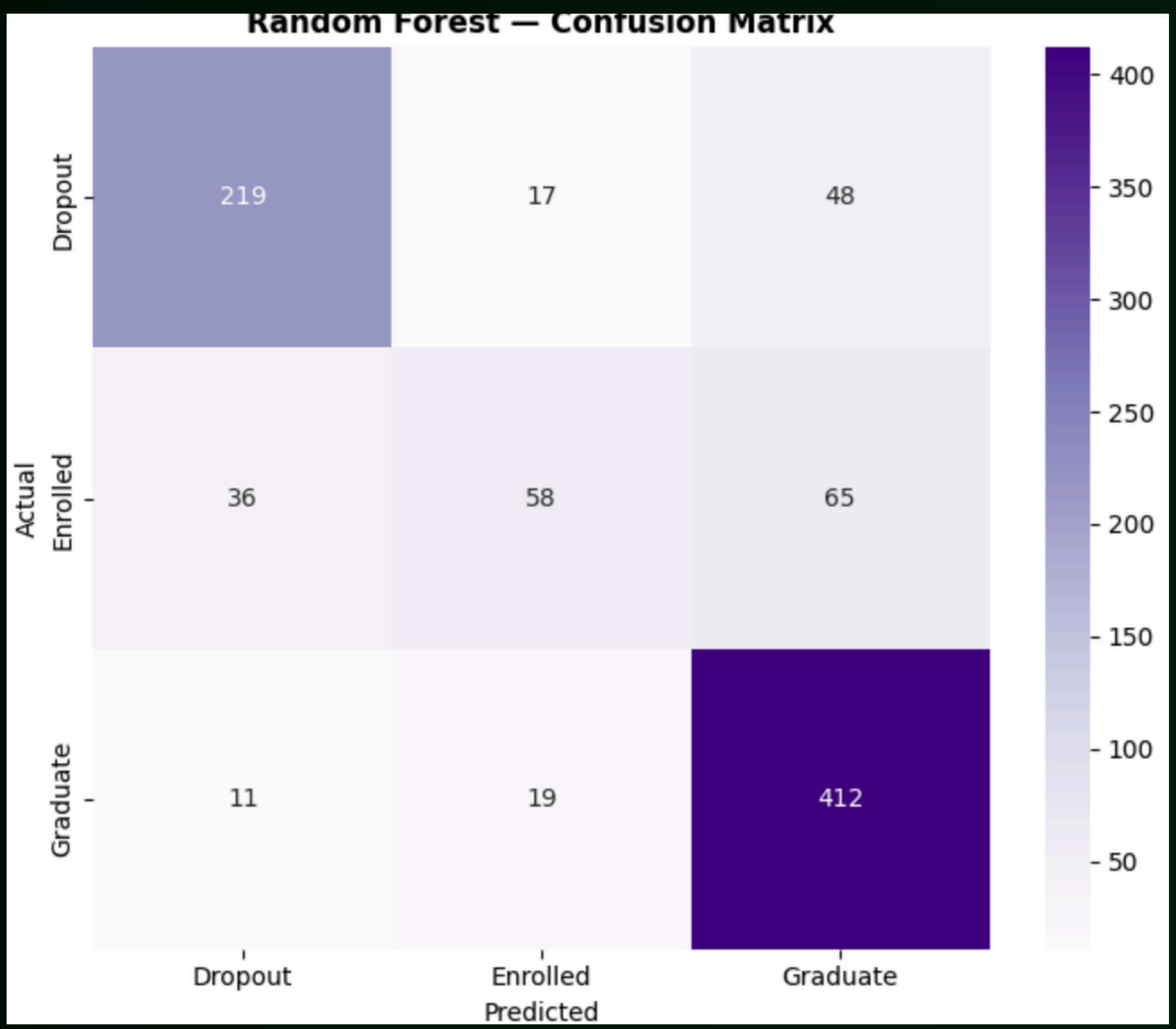
=== Random Forest ===

Accuracy: 0.779

	precision	recall	f1-score	support
Dropout	0.82	0.77	0.80	284
Enrolled	0.62	0.36	0.46	159
Graduate	0.78	0.93	0.85	442
accuracy			0.78	885
macro avg	0.74	0.69	0.70	885
weighted avg	0.77	0.78	0.76	885

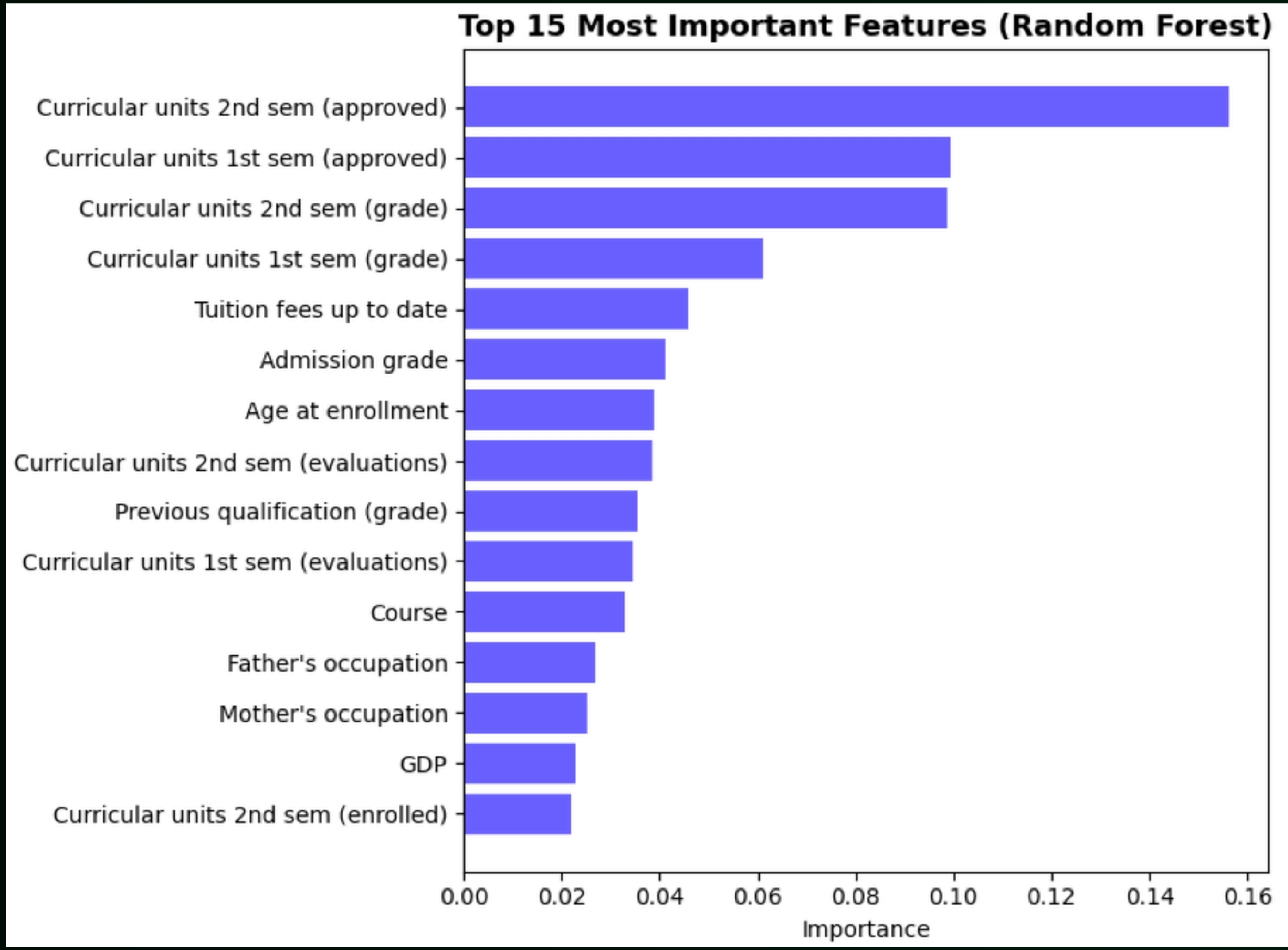
Confusion Matrix: Where Random Forest Gets It Right (and Wrong)

of 284 actual Dropouts, the model correctly caught 219 (75%), but misclassified 48 as Graduates. Of 442 actual Graduates, it correctly identified 412 (94%) — the model's strongest performance by far. Of 159 actual Enrolled students, it only got 58 right (31%), with most of the rest split between being mistaken for Dropouts (36) or Graduates (65). The takeaway: the model is genuinely useful at flagging at-risk dropouts and confidently identifying students on track to graduate, but it can't reliably tell where a currently-enrolled student is headed yet — which actually lines up with real life, since their fate hasn't played out yet.



What Drives the Predictions:
Feature Importance

the model confirms what we found earlier." The top features are almost entirely academic performance metrics — 2nd semester approved units, 1st semester approved units, and grades from both semesters dominate the chart, with tuition payment status (Tuition fees up to date) right behind them. This directly validates your earlier EDA finding that students who fall behind early (especially financially) are the ones most likely to drop out. You can literally say: "Our model independently rediscovered the same patterns we found by hand in our EDA — early academic performance and financial standing are the strongest predictors of student outcome.



Next Steps

Incorporating a “Risk Index”: Building a risk score that combines multiple factors (e.g. GPA, financial aid status, etc.) into one index and flagging any score that goes below a certain threshold.

Longitudinal Validation: Testing our model’s across different institution types (e.g. community college vs. four-year institution, public vs. private, etc.) to see how our predictions hold up across schools.

Works Cited

Delogu, M., Lagravinese, R., Paolini, D., & Resce, G. (2024). Predicting dropout from higher education: Evidence from Italy. *Economic Modelling, 130,* 106583. https://doi.org/10.1016/j.econmod.2023.106583

THANK YOU